



Name: \_\_\_\_\_ Cohort/Batch: \_\_\_\_\_ Date: \_\_\_\_\_ Score: \_\_\_\_\_

MICROSERVICES PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Architecture

TOTAL: 10 QUESTIONS

**Q1** What is Microservices and what problem in Architecture does it solve?

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**Q6** How does Microservices handle reliability and high availability?

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**Q2** What are the core components or concepts in Microservices?

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**Q7** What observability does Microservices provide out of the box?

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**Q3** How does Microservices compare to its main alternatives?

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**Q8** What are common performance bottlenecks in Microservices?

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**Q4** How do you get started with Microservices for a new project?

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**Q9** What security best practices apply when running Microservices?

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**Q5** What configuration options most affect Microservices's behavior in production?

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**Q10** What mistakes do teams commonly make when adopting Microservices?

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ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Microservices is a tool or platform that

Mitigation

Microservices is a tool or platform that automates or simplifies a specific concern in the Architecture domain, reducing manual effort and operational complexity for engineering teams.

A6 Microservices provides HA through leader election, Observability

Microservices provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 Microservices has key building blocks that work

Primitives

Microservices has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 Microservices exposes metrics (Prometheus endpoint or Information

Microservices exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A3 Microservices trades off simplicity against feature richness

Refinement

Microservices trades off simplicity against feature richness differently than its alternatives. Choose Microservices when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A4 Install Microservices via its package manager or

Gotchas

Install Microservices via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run Microservices with the principle of least

Assessment

Run Microservices with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency

Configuration

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the documentation and

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.